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**LEVEL:** 200LEVEL

**COURSE CODE:** COS 202

## QUESTIONS

**1. Write and run a C++ code that finds the tallest boy and girl in COS 202 class**

**2. Write a C++ code to compute basic arithmetic operations**

### 1. Code Output

| main.cpp                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              | Output                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <pre>1 #include &lt;iostream&gt; 2 #include &lt;string&gt; 3 4 using namespace std; 5 //Program to find the tallest Boy and girl in a class 6 int main(){ 7     string name; 8     int n; // number of students 9     char gender; 10    string TallestGirl, TallestBoy; 11    double TallestBoyH = 0, TallestGirlH = 0, Height; 12 13    cout &lt;&lt; "Please Input the number of students: "; 14    cin &gt;&gt; n; 15 16    for (int i = 0; i &lt; n; i++){ 17        cout &lt;&lt; "Input student name: "; 18        getline(cin &gt;&gt; ws, name); 19 20        cout &lt;&lt; "Input student gender(m/f): "; 21        cin &gt;&gt; gender; 22 23        cout &lt;&lt; "Input student height in cm (1 - 1000): "; 24        cin &gt;&gt; Height; 25    }</pre> | <pre>Please Input the number of students: 5 Input student name: Nifemi Input student gender(m/f): m Input student height in cm (1 - 1000): 500   Input student name: Felicia Input student gender(m/f): f Input student height in cm (1 - 1000): 300   Input student name: James Input student gender(m/f): m Input student height in cm (1 - 1000): 500   Input student name: John Input student gender(m/f): m Input student height in cm (1 - 1000): 700   Input student name: Favout Input student gender(m/f): f Input student height in cm (1 - 1000): 700   The tallest boy is John (700) The tallest girl is Favout (700)</pre> |

## C++ Source Code: Tallest Boy and Girl Finder

```
#include <iostream>
```

```
#include <string>
```

```
using namespace std;
```

```
//Program to find the tallest Boy and girl in a class
```

```
int main(){
```

```
string name;

int n; // number of students

char gender;

string TallestGirl, TallestBoy;

double TallestBoyH = 0, TallestGirlH = 0, Height;


cout << "Please Input the number of students: ";
cin >> n;


for (int i = 0; i < n; i++){
    cout << "Input student name: ";
    getline(cin >> ws, name);


    cout << "Input student gender (m/f): ";
    cin >> gender;


    cout << "Input student height in cm (1 - 1000): ";
    cin >> Height;


    cout << endl;


    if (gender == 'm'){
        if (Height > TallestBoyH){
            TallestBoyH = Height;
            TallestBoy = name;
        }
    }

    else if (gender == 'f'){
```

```
        if (Height > TallestGirlH){
            TallestGirlH = Height;
            TallestGirl = name;
        }
    }

    else{
        cout<< "Error: An error occured";
    }
}

    cout << "The tallest boy is " << TallestBoy << " (" <<
TallestBoyH << ")"<< endl;

    cout << "The tallest girl is " << TallestGirl << " (" <<
TallestGirlH << ")" << endl;

}
```

## 2. Code Output

```
main.cpp  [Icons]  Share  Run  Output  Clear

1  #include <iostream>
2  using namespace std;
3
4  // Program to Calculate Basic Arithmetic operates
5
6  int main(){
7      double a, b;
8      char op;
9
10     cout << "Input a: ";
11     cin >> a;
12     cout << "Input b: ";
13     cin >> b;
14     cout << "Input operator (*, +, -, /): ";
15     cin >> op;
16
17     switch(op){
18         case '+':
19             cout << a << " + " << b << " = " << a + b << endl;
20             break;
21         case '-':
22             cout << a << " - " << b << " = " << a - b << endl;
23             break;
24         case '/':
25             if (b != 0){
```

Input a: 2  
Input b: 5  
Input operator (\*, +, -, /): \*  
2 \* 5 = 10

=== Code Execution Successful ===

## C++ Source Code: Basic arithmetic operations computation

```
#include <iostream>

using namespace std;

// Program to Calculate Basic Arithmetic operates

int main(){
    double a, b;
    char op;

    cout << "Input a: ";
    cin >> a;
    cout << "Input b: ";
    cin >> b;
    cout << "Input operator (*, +, -, /): ";
    cin >> op;

    switch(op){
        case '+':
            cout << a << " + " << b << " = " << a + b << endl;
            break;
        case '-':
            cout << a << " - " << b << " = " << a - b << endl;
            break;
        case '/':
            if (b != 0){
                cout << a << " / " << b << " = " << a/b << endl;
            }
    }
```

```
    }
    else{
        cout << "Error: Division by zero returns
undefined" << endl;
    }
    break;

    case '*':
        cout << a << " * " << b << " = " << a * b << endl;
        break;

    default:
        cout << "Error: Invalid operator Inputed" << endl;
}
return 0;
}
```